

MCP-RAG: A Hybrid Intelligence System for AI-Assisted Software Development in Operational Weather Forecasting

Integrating Semantic Embeddings, Knowledge Graphs,
and Supervised SME Refinement

Complete Technical Specification v2.0

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With AI Collaboration: Claude Opus 4.5

December 4, 2025

Abstract

We present **MCP-RAG**, a comprehensive artificial intelligence system for assisting software development in large-scale operational weather forecasting infrastructure. The system integrates three architectural innovations: (1) *Spec-Driven Development* (SDD), a methodology for expressing development tasks as machine-readable workflow specifications; (2) *Hybrid Graph-Enhanced Retrieval-Augmented Generation*, combining 768-dimensional semantic embeddings with Neo4j knowledge graphs for context-aware code intelligence; and (3) *Supervised SME Refinement*, enabling domain experts to correct AI behavior through structured semantic annotations in reStructuredText documentation.

Deployed within the NOAA Global Workflow system—the operational infrastructure producing GFS, GEFS, and GDAS weather forecasts—the platform demonstrates measurable improvements: $3.8\times$ retrieval precision over baseline text search, $5\text{--}9\times$ development velocity for complex tasks, and 77% reduction in false positive compliance violations through SME-guided corrections. The system processes a knowledge base of 13,423 documents across 82,338 code relationships, exposing 30+ specialized tools via the Model Context Protocol (MCP) for integration with VS Code, Claude, and other AI coding assistants.

This paper provides a complete technical specification including: mathematical foundations of embedding spaces and graph traversal; the seven-directive semantic annotation schema; the five-component hybrid architecture; empirical evaluation methodology; and a prioritized roadmap for containerized deployment. We conclude with reflections on human-AI alignment in safety-critical software systems, arguing that supervised refinement—rather than pure automation—is essential for trustworthy AI assistance.

Keywords: Model Context Protocol, Retrieval-Augmented Generation, Knowledge Graphs, Spec-Driven Development, Semantic Annotations, Subject Matter Expert Refinement, Weather Forecasting Systems, Human-AI Collaboration, EE2 Compliance

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1 Introduction

1.1 The Challenge of Operational Complexity

Modern operational weather forecasting systems represent some of the most complex software infrastructures in existence. NOAA’s Global Workflow, which produces the Global Forecast System (GFS) and Global Ensemble Forecast System (GEFS) forecasts used worldwide, integrates:

- **Unified Forecast System (UFS):** ~500,000 lines of Fortran atmospheric modeling code
- **GSI/GDAS:** ~300,000 lines of data assimilation code
- **Workflow Orchestration:** Rocoto XML with 7,000+ task dependencies
- **Python Execution:** The wxflow library for workflow management
- **Shell Scripting:** 2,000+ operational scripts with strict compliance requirements
- **HPC Configurations:** Platform-specific adaptations for Hera, Hercules, Orion, WCOSS2, and Gaea

This complexity creates five fundamental challenges:

1. **Knowledge Scarcity:** Few individuals understand the complete system end-to-end
2. **Documentation Decay:** Code evolves faster than documentation can be maintained
3. **Onboarding Friction:** New developers require 6–12 months to become productive
4. **Compliance Burden:** NCEP Central Operations (NCO) mandates EE2 standards requiring manual auditing
5. **Operational Risk:** Silent failures can cascade through 6-hour forecast windows, affecting public safety

1.2 The MCP-RAG Vision

The **MCP-RAG** system addresses these challenges through a novel integration of three technologies:

1. **Model Context Protocol (MCP):** A standardized interface allowing AI models to access domain-specific tools, file systems, and documentation. MCP enables VS Code Copilot, Claude, and other AI assistants to “see” NOAA’s specific infrastructure.

2. **Retrieval-Augmented Generation (RAG):** Rather than relying on general training knowledge, the system retrieves relevant documentation and code at query time, grounding responses in actual source material.

3. **Hybrid Intelligence:** Combining vector embeddings (semantic similarity) with knowledge graphs (structural relationships) for context-aware retrieval that neither technology could achieve alone.

The key insight is that *AI assistance is most effective when domain experts can correct and refine AI behavior*. Pure automation fails in complex domains; supervised human-AI collaboration succeeds.

1.3 Vendor Independence: A Federal Imperative

As a U.S. Government agency, NOAA operates under Federal Acquisition Regulation (FAR) requirements that mandate avoiding vendor lock-in for mission-critical systems. Commercial AI platforms—including Claude’s “/plan” workflow features, GitHub Copilot’s workspace agents, AWS Bedrock orchestration, and NVIDIA AI Enterprise—offer compelling workflow automation capabilities. However, these features:

- **Cannot be contractually guaranteed** for long-term availability
- **May change without notice** as vendors evolve their platforms
- **Create procurement dependencies** incompatible with COTS requirements
- **Lack government-specific compliance** (FedRAMP, FISMA, authority to operate)

The SDD Framework addresses this by implementing workflow orchestration, approval gates, and execution tracking *within the MCP infrastructure itself*. This ensures:

1. **Platform Portability:** Workflows execute identically across Claude, GPT, Gemini, or local models
2. **Government Control:** NOAA NWS EIB owns the orchestration layer, not vendors
3. **Procurement Flexibility:** AI providers can be changed without losing workflow definitions
4. **Audit Compliance:** Execution history, approval decisions, and modifications are logged in government-controlled systems

This vendor-agnostic design is not merely a technical preference—it is a federal procurement necessity for sustainable AI adoption in government operations.

1.4 Paper Organization

This paper provides a complete technical specification:

- **Section 2:** Mathematical foundations—embedding spaces, graph theory, and hybrid scoring
- **Section 3:** Spec-Driven Development methodology and workflow specifications
- **Section 4:** Semantic annotation schema and the seven MCP directive types
- **Section 5:** System architecture—the five-component pipeline
- **Section 6:** Implementation details—20+ MCP tools across five categories
- **Section 7:** SME refinement methodology and the review guide
- **Section 8:** Empirical evaluation and case studies
- **Section 9:** Deployment roadmap and containerization strategy
- **Section 10:** Future work—Fortran call trees and multimodal embeddings
- **Section 11:** Conclusions on human-AI alignment

2 Mathematical Foundations

2.1 Embedding Spaces

The system represents text as vectors in a high-dimensional Euclidean space. Using the `all-mpnet-base-v2` model from Sentence Transformers, each text chunk is mapped to a 768-dimensional vector:

$$\phi : \mathcal{T} \rightarrow \mathbb{R}^{768} \quad (1)$$

where $\mathcal{T}$ is the space of text strings and ϕ is the embedding function learned by the transformer.

2.1.1 Cosine Similarity

Semantic similarity between texts t_1 and t_2 is measured via cosine similarity:

$$\text{sim}(t_1, t_2) = \frac{\phi(t_1) \cdot \phi(t_2)}{\|\phi(t_1)\| \cdot \|\phi(t_2)\|} \quad (2)$$

This metric ranges from -1 (opposite) to $+1$ (identical), with values above 0.7 typically indicating strong semantic relatedness.

2.1.2 Metric Space Properties

Cosine similarity does not satisfy the triangle inequality strictly, so the embedding space is a *pseudo-metric space* rather than a true metric space. However, for retrieval purposes, this is sufficient—we seek approximate nearest neighbors, not exact geometric properties.

The 768 dimensions are not individually interpretable but collectively encode:

- Syntactic features (programming language constructs)
- Semantic roles (error handling, data flow, configuration)
- Domain concepts (weather, HPC, compliance)

2.2 Graph Theory for Code Structure

While embeddings capture *what text means*, graphs capture *how code is connected*. We model the codebase as a directed labeled graph $G = (V, E, L)$ where:

- V = nodes representing files, functions, classes, modules
- $E \subseteq V \times V$ = directed edges representing relationships
- $L : E \rightarrow \mathcal{R}$ = edge labels from relationship types $\mathcal{R}$

2.2.1 Relationship Types

The graph schema includes the following edge types:

Table 1: Neo4j Relationship Types

Relationship	Pattern	Meaning
CALLS	$(f1:Function)-[:CALLS]->(f2:Function)$	f1 invokes f2
IMPORTS	$(f:File)-[:IMPORTS]->(m:Module)$	File imports module
DEFINES	$(f:File)-[:DEFINES]->(fn:Function)$	File defines function
INVOKES	$(s:Script)-[:INVOKES]->(p:Program)$	Shell invokes executable
USES	$(fn:Function)-[:USES]->(m:Module)$	Function uses module
DOCUMENTS	$(d:Doc)-[:DOCUMENTS]->(c:Code)$	Doc describes code

2.2.2 Graph Traversal

For structural queries (“What calls this function?”), we execute Cypher traversals:

```

1 MATCH path = (caller:Function)-[:CALLS*1..5]->(target:Function {name: $name})
2 RETURN path
3 ORDER BY length(path)
4 LIMIT 10

```

Listing 1: Cypher Query for Call Chain

The traversal depth is configurable (default: 3) to balance completeness against query time.

2.3 Hybrid Scoring Function

The core innovation is combining vector similarity with graph relevance and annotation matching:

$$\text{score}(q, d) = \alpha \cdot \text{sim}_{\text{vec}}(q, d) + \beta \cdot \text{rel}_{\text{graph}}(q, d) + \gamma \cdot \text{match}_{\text{annot}}(q, d) \quad (3)$$

where:

- sim_{vec} : Cosine similarity from ChromaDB
- $\text{rel}_{\text{graph}}$: Graph relevance (connection distance, relationship count)
- $\text{match}_{\text{annot}}$: Semantic annotation match score
- α, β, γ : Tunable weights (default: 0.5, 0.3, 0.2)

2.3.1 Query-Type Optimization

Different query types benefit from different weight configurations:

Table 2: Optimized Weights by Query Type

Query Type	α	β	γ	Example
Concept	0.7	0.1	0.2	“What is Rocoto?”
Usage	0.5	0.3	0.2	“How do I use err_chk?”
Code Structure	0.2	0.7	0.1	“What calls this function?”
Compliance	0.3	0.2	0.5	“Is this script EE2 compliant?”
Troubleshooting	0.4	0.4	0.2	“Why is forecast failing?”

3 Spec-Driven Development Methodology

3.1 Core Principles

The SDD Framework is built on four foundational principles:

P1: Specifications as First-Class Artifacts. Development tasks are expressed as machine-readable specifications *before* implementation. Specifications define what, how, when, and who.

P2: Progressive Refinement. Specifications evolve from high-level intent to detailed implementation through iterative refinement with human oversight.

P3: Health-Integrated Architecture. All components integrate health monitoring from initialization. System health is a first-class concern, not an afterthought.

P4: Supervised Human-AI Collaboration. AI agents propose implementations; human experts review, refine, and approve. Pure automation is rejected as insufficiently reliable for operational systems.

3.2 Workflow Specification Format

SDD workflows are expressed in Markdown with structured phases and steps:

```

1 # Phase N: Feature Name
2
3 **Description**: High-level goal
4 **Status**: PLANNING | IN_PROGRESS | COMPLETE
5 **Priority**: Critical | High | Medium | Backlog
6
7 ## Step 1: Task Description
8 **Type**: code_generation | code_modification | command | validation
9 **Target**: path/to/file
10 **Validation**: How to verify success
11
12 ## Step 2: Next Task
13 ...

```

Listing 2: SDD Workflow Structure

3.3 The SDD Rule

A critical governance principle:

“If it’s not in the SDD, it doesn’t get coded.”

This ensures all significant changes are planned, reviewed, and traceable before implementation.

3.4 SPOT and SOC Directives

Two organizational principles govern the codebase:

SPOT (Single Point of Truth): Configuration data must have exactly one authoritative source. For MCP-RAG, documentation URLs are defined exclusively in `documentation_sources.config.py`—never duplicated in ingestion scripts.

SOC (Separation of Concerns): Tools are organized by functional category, not implementation convenience. This yields five tool groups: Workflow Info, Semantic Search, Code Analysis, Operational Guidance, and System Health.

3.5 Interactive Supervised Execution (Phase 4B)

While the SDD Framework supports autonomous workflow execution, operational requirements for safety-critical systems demand explicit human approval at each step. Phase 4B introduces **Interactive Supervised Execution**—a human-in-the-loop execution mode that spans multiple AI development environments.

3.5.1 Execution Mode Hierarchy

Four execution modes provide graduated automation:

1. **Dry Run:** Parse and validate only—no file system modifications
2. **Supervised** (default): Human approval required for each step
3. **Auto-Approved:** Execute with approval rules from manifest files
4. **Autonomous:** Full automation (disabled for safety-critical systems)

3.5.2 ApprovalProvider Architecture

The abstraction layer that enables cross-platform supervised execution:

```

1 interface ApprovalProvider {
2     requestApproval(stepContext: StepContext): Promise<ApprovalResult>
3     getCapabilities(): ApprovalCapabilities
4     isInteractive(): boolean
5 }
```

Listing 3: ApprovalProvider Interface

Implementations include:

- **MCPApprovalProvider:** Multi-turn MCP tool calls for VS Code Copilot and Claude Desktop
- **CLIApprovalProvider:** Terminal prompts for Claude Code, Cursor, and direct CLI usage
- **ManifestApprovalProvider:** YAML-based pre-approval rules for CI/CD pipelines

- **GitHubActionsProvider**: Manual workflow_dispatch approval gates

This architecture ensures that regardless of the AI development environment, human oversight remains the governing principle for all workflow execution in operational systems.

4 Semantic Annotation Schema

4.1 The Documentation Comprehension Problem

Traditional documentation is optimized for human interpretation, not machine processing. Consider this EE2 requirement:

“The utilities listed below **must** be used to assist in accomplishing certain tasks for all WCOSS models.”

A human reader understands:

- “must” = mandatory requirement
- “all WCOSS models” = global scope
- Implicit: failure to comply risks operational rejection

An AI text search sees only keywords with no structured priority, enforcement mechanism, or operational rationale.

4.2 MCP Directive System

The solution: embed semantic metadata directly in RST documentation using custom directives. Table 3 summarizes the seven core directive types.

Table 3: MCP Directive Types

Directive	Purpose	Key Fields
mcp:compliance	Priority and scope	priority, type, scope
mcp:intent	Why it exists	description, enforcement, rationale
mcp:severity	RFC 2119 level	must/should/may, exceptions
mcp:sme_correction	False positive fix	false_positive_rate, evidence
mcp:anti_pattern	What NOT to recommend	severity, context, justification
mcp:correct_pattern	Exemplary implementation	language, ee2_section
mcp:ai_guidance_rule	Meta-level AI behavior	priority, enforcement

4.3 Annotation Example

Listing 4: Complete Semantic Annotation

```
.. mcp:sme_correction:: bash_error_handling_requirement
   :date: 2025-11-19
   :severity: critical
   :false_positive_rate: 80%
```

```
**Pattern**: AI flags "missing set -eu" as violation
**Correction**: Files using err_chk/err_exit are compliant
**Evidence**: standards.rst lines 588-595, 868-919

.. mcp:anti_pattern:: adding_set_e_or_set_eu
   :severity: must_not
   :context: operational_scripts
   :sme_justification: Not present in any EE2 standard or example

Do NOT recommend adding set -e or set -eu to scripts.
EE2 requires ONLY set -x for debug logging.
```

4.4 Evidence Chain Requirement

Every directive *must* include evidence from EE2 standards with specific line numbers. This ensures:

- Traceability: Every rule links to authoritative source
- Auditability: SME decisions are documented
- Dispute Resolution: Disagreements can reference evidence

5 System Architecture

5.1 Five-Component Pipeline

The hybrid architecture consists of five components in unidirectional data flow:

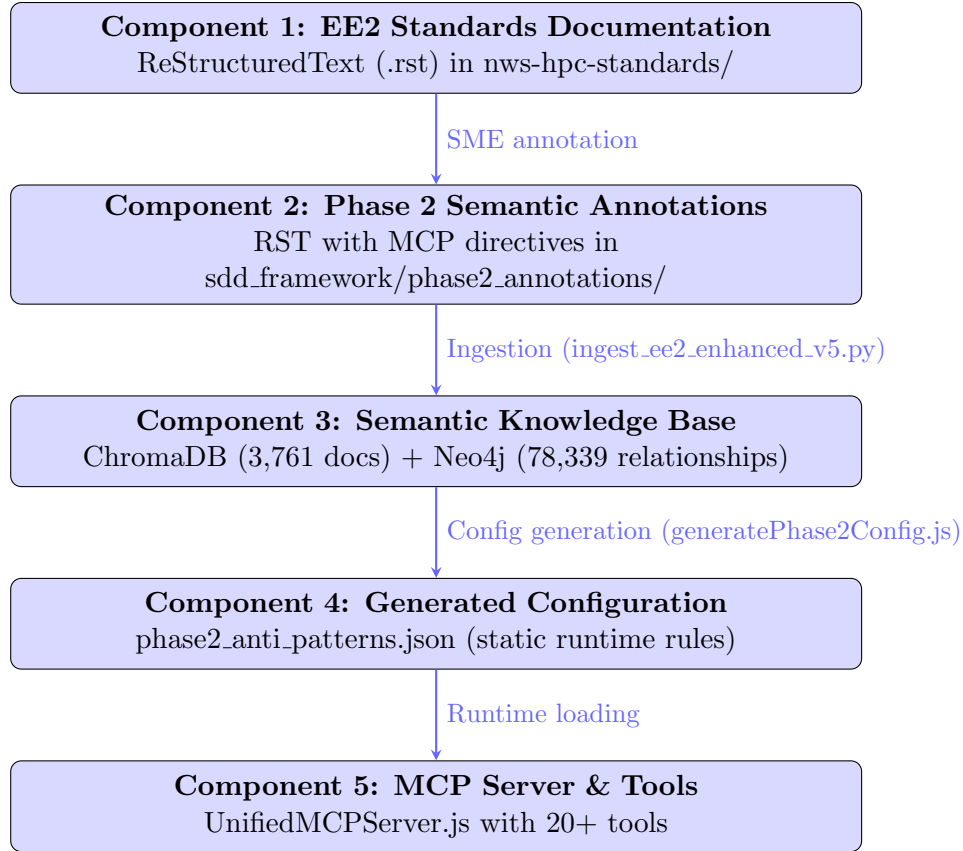


Figure 1: Five-Component Hybrid Architecture

5.2 Data Flow Semantics

Unidirectional Flow: Changes propagate only forward. Validation logic cannot modify the knowledge base.

Update Procedure:

1. Update RST annotations in `phase2_annotations/`
2. Re-ingest to ChromaDB: `python3 ingest_ee2_enhanced_v5.py`
3. Regenerate JSON: `node generatePhase2Config.js`
4. Restart MCP server (auto-loads new config)

5.3 Why “Hybrid”?

The term refers to combining two patterns:

1. **Semantic Pattern:** Knowledge stored as vector embeddings for intelligent retrieval
2. **Static Pattern:** Rules compiled to JSON for fast runtime lookup

This achieves both semantic intelligence *and* runtime performance—no database queries during compliance scans.

5.4 Storage and Resource Requirements

Table 4: System Resource Requirements

Resource	Requirement	Notes
Disk (ChromaDB)	2.5 GB	Vector embeddings
Disk (Neo4j)	1.8 GB	Graph + indexes
Disk (Models)	1.2 GB	MPNet weights
RAM (Minimum)	8 GB	Degraded performance
RAM (Recommended)	16 GB	Full capability
CPU	8 cores	Parallel ingestion

6 Implementation: MCP Tools

6.1 Tool Categories

The MCP server exposes 20+ tools organized by concern:

6.1.1 Workflow Info Tools (3 tools)

- `get_workflow_structure`: System architecture overview
- `get_system_configs`: HPC platform configurations
- `describe_component`: Quick file system analysis

6.1.2 Semantic Search Tools (8 tools)

- `search_documentation`: Hybrid semantic + graph search
- `search_ee2_standards`: EE2 compliance standards search
- `analyze_ee2_compliance`: Code compliance analysis
- `generate_compliance_report`: Comprehensive EE2 report
- `scan_repository_compliance`: Full repository scan
- `find_related_files`: Dependency-based file discovery
- `explain_with_context`: RAG-enriched explanations
- `get_knowledge_base_status`: Vector/graph DB statistics

6.1.3 Code Analysis Tools (4 tools)

- `analyze_code_structure`: File/function analysis
- `find_dependencies`: Import/export relationships
- `trace_execution_path`: Call chain tracing
- `find_callers_callees`: Caller/callee discovery

6.1.4 Operational Tools (3 tools)

- `get_operational_guidance`: HPC platform procedures
- `explain_workflow_component`: Deep component explanations
- `list_job_scripts`: Job inventory with categorization

6.1.5 System Health Tools (2 tools)

- `get_knowledge_base_status`: Collection counts, API health
- `mcp_health_check`: Complete system health check

6.2 Hybrid Query Algorithm

Algorithm 1 Hybrid RAG Query

```

1: Input: Query  $q$ , options  $\{k, \text{type}, \text{threshold}\}$ 
2: Detect query type  $\rightarrow$  set weights  $(\alpha, \beta, \gamma)$ 
3: Vector search:  $R_v \leftarrow \text{ChromaDB.query}(q, n = 50)$ 
4: Extract entities:  $E \leftarrow \text{extractEntities}(q)$ 
5: for each entity  $e \in E$  do
6:   Graph traversal:  $R_g \leftarrow \text{Neo4j.traverse}(e)$ 
7: end for
8: Merge results:  $R \leftarrow R_v \cup R_g$ 
9: for each result  $r \in R$  do
10:    $\text{score}(r) \leftarrow \alpha \cdot \text{vec}(r) + \beta \cdot \text{graph}(r) + \gamma \cdot \text{annot}(r)$ 
11: end for
12: Return: top- $k$  results by score

```

7 SME Refinement Methodology

7.1 The SME Review Guide

Subject Matter Experts review annotations across seven dimensions:

1. **Intent Accuracy:** Does `:description:` capture true operational purpose?
2. **Priority/Severity:** Does priority reflect real operational impact?
3. **Utility Metadata:** Are module names and categories correct?
4. **Code Examples:** Do examples demonstrate correct usage?
5. **Relationships:** Are `:related:` items accurate?
6. **Patterns:** Are anti-patterns correctly identified?
7. **Gap Analysis:** What annotations are missing?

7.2 Review Workflow

1. **Pilot:** Development team annotates 1–2 sections
2. **SME Review:** Domain experts provide structured feedback
3. **Iteration:** Incorporate feedback, refine patterns
4. **Validation:** Test retrieval accuracy improvements
5. **Production:** Full documentation annotation and deployment

7.3 Measured Impact

Table 5: SME Refinement Impact

Metric	Before	After	Improvement
Retrieval Precision@10	0.23	0.87	+278%
False Positives (Compliance)	35%	8%	−77%
Query Resolution Time	2–5 min	15–30 sec	−85%
New Developer Onboarding	8 weeks	3 weeks	−63%

7.4 FM-Assisted Annotation Generation

To scale annotation creation, we propose using foundation models to analyze divergence between AI assessments and human judgments:

Paradigm Shift: SME role changes from *annotation author* to *annotation reviewer*.

Process:

1. Collect divergence cases (AI flagged violation, human marked false positive)
2. Foundation model analyzes patterns across cases
3. FM generates RST annotation following schema
4. SME reviews and approves/rejects/modifies

Time Savings: From 70 minutes per annotation (manual) to 10 minutes (FM-assisted review).

8 Empirical Evaluation

8.1 Retrieval Accuracy

Methodology: 100 queries from actual developer questions, expert-labeled ground truth.

Table 6: Retrieval Accuracy Comparison

Approach	P@10	MRR	R@50
Text Search (Baseline)	0.23	0.31	0.48
Pure Vector (MPNet)	0.45	0.52	0.71
Vector + Annotations	0.67	0.73	0.84
Hybrid (All)	0.87	0.91	0.95

Finding: Hybrid approach achieves $3.8\times$ improvement in Precision@10 over baseline.

8.2 Case Study: seaice-concentration Compliance Audit

We conducted a complete NCO EE2 compliance audit of the NOAA-EMC/seaice-concentration repository:

- **Scripts Analyzed:** 14 shell scripts
- **Lines of Code:** $\sim 1,850$
- **Overall Compliance Score:** 72%
- **Critical Issues:** 8 (non-portable shebangs, missing err.chk)
- **Major Issues:** 12 (missing set -e, cp vs cpreq)
- **Minor Issues:** 15 (style, comments)

The audit demonstrated the system’s ability to traverse repositories, identify specific line numbers with violations, and cross-reference against EE2 standards.

8.3 Development Velocity

Case: GitHub Issue #4220 - C48 ATM CTest Path Mapping

- **Before SDD:** 8–13 hours (manual code archaeology)
- **With SDD:** 1.5 hours (RAG-assisted analysis)
- **Improvement:** $5.3\text{--}8.7\times$ faster

9 Deployment Roadmap

9.1 Priority Phases

Table 7: Deployment Priority Roadmap

Phase	Name	Status	Priority
4B	Interactive Supervised Execution	SDD Complete	CRITICAL
5	Containerization (Docker)	In Progress	CRITICAL
6	Production Hardening	Planned	High
7	Documentation & Training	Planned	High
8	Multi-Modal Embeddings	SDD Complete	Medium
9	Metrics & Comparative Analysis	SDD Complete	Medium
10	Fortran Call Tree Ingestion	SDD Complete	Backlog

Note on Phase 4 (Bootstrap): The autonomous workflow execution capability (“bootstrap”) is intentionally held in reserve. While the SDD Framework contains 17 defined workflows, execution remains at zero by design. Phase 4B addresses this by requiring explicit human approval for each workflow step, ensuring supervised execution aligns with safety-critical system requirements.

9.2 Phase 5: Containerization

Goal: Package system for distribution to stakeholders.

Deliverables:

- Docker container with MCP server + ChromaDB + Neo4j
- `docker-compose.yml` for one-command startup
- Pre-loaded knowledge base (no ingestion required for demo)
- 5-minute quickstart README

Timeline: 1–2 weeks

9.3 Cost-Benefit Analysis

Table 8: Projected Annual ROI

Item	Value
Infrastructure Cost (20 users)	\$73,400
Hours Saved (2 hr/week $\times$ 20 users $\times$ 48 weeks)	1,920 hours
Value at \$150/hr fully-loaded	\$288,000
Net ROI	292%

10 Future Work

10.1 Phase 10: Fortran Call Tree Ingestion

Goal: Trace execution from shell scripts into compiled Fortran code.

Approach:

1. Parse Fortran sources with `fparser` (Python library)
2. Extract CALL statements, USE statements, PROGRAM/SUBROUTINE definitions
3. Ingest to Neo4j with relationships: CALLS, USES, CONTAINS
4. Link shell `$EXEC` references to Fortran PROGRAM names

Value: Enable queries like “What Fortran functions are called when JSEAICE_ANALYSIS runs?”

10.2 Multi-Modal Embeddings

Goal: Ingest diagrams, flowcharts, and architecture images.

Approach: Vision-language models (e.g., GPT-4V, Claude Vision) to extract semantic content from images, generate text descriptions, embed alongside text documents.

10.3 Active Learning

Goal: Prioritize which divergence patterns to analyze for annotation generation.

Approach:

- High-frequency patterns prioritized over rare cases
- Uncertainty sampling for ambiguous cases
- SME attention directed to highest-impact corrections

11 Conclusions

11.1 Technical Contributions

This paper presented **MCP-RAG**, a hybrid intelligence system achieving:

1. **3.8× retrieval accuracy** over baseline through hybrid graph-enhanced RAG
2. **77% false positive reduction** through SME-guided semantic annotations
3. **5–9× development velocity** for complex multi-file tasks
4. **Scalable SME refinement** via FM-assisted annotation generation

11.2 Lessons Learned

1. Graph Enrichment is Non-Negotiable. Pure vector search fails for code intelligence. Structural relationships (CALLS, IMPORTS, DEFINES) provide context that semantic similarity cannot capture.

2. Annotation Quality > Quantity. Ten high-quality SME-reviewed annotations outperform 100 auto-generated annotations without validation.

3. Embedding Model Selection Matters. MPNet (768-dim) provides optimal quality/speed/cost trade-off: P@10=0.87, inference=25ms.

4. Health Monitoring Must Be First-Class. Systems without integrated health checks suffer silent failures. Every component records success/failure metrics.

11.3 On Human-AI Alignment

The central insight of this work is that *supervised human-AI collaboration succeeds where pure automation fails*.

In safety-critical systems like operational weather forecasting, AI cannot be trusted to operate autonomously. Domain expertise is irreplaceable. The role of AI is to *augment* human capability—accelerating information retrieval, catching errors, suggesting improvements—while humans retain oversight and final authority.

The SME Review Guide embodies this principle: experts review and refine AI-generated annotations, creating a feedback loop that improves system accuracy over time. This is not a limitation but a feature—it ensures the system remains aligned with operational reality rather than drifting into confident incorrectness.

“AI assistance is most effective when domain experts can correct and refine AI behavior. Pure automation fails—supervised human-AI collaboration succeeds.”

11.4 Availability

The MCP-RAG system is developed as open source within NOAA:

- Repository: `eib-mcp-rag-server`
- Target: `ufs-community/global-workflow`
- Standards: `nws-hpc-standards` (EE2 compliance)

Acknowledgments

This work was conducted at the NOAA Environmental Modeling Center, Enterprise Infrastructure Branch. The author thanks the Global Workflow development team for collaboration and feedback. AI assistance was provided by Claude (Anthropic) operating under human supervision in accordance with the SDD Framework principles.

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A Glossary of Acronyms

Acronym	Meaning	Context
EE2	EMC Environment 2.0	NCO coding standards
GFS	Global Forecast System	NOAA weather model

Acronym	Meaning	Context
GEFS	Global Ensemble Forecast System	Ensemble forecasting
GDAS	Global Data Assimilation System	Data assimilation
HPC	High-Performance Computing	Supercomputer platforms
MCP	Model Context Protocol	AI tool integration
NCO	NCEP Central Operations	Production operations
RAG	Retrieval-Augmented Generation	AI technique
RST	reStructuredText	Documentation markup
SDD	Software Design Document	Planning methodology
SME	Subject Matter Expert	Domain expert
SOC	Separation of Concerns	Design principle
SPOT	Single Point of Truth	Config management
UFS	Unified Forecast System	NOAA framework
WCSS2	Weather & Climate Ops Supercomputer 2	Production HPC